

Assignment 12

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Objective

The main goal of this project was to:

1. Build a small CNN classifier trained on the CIFAR-10 dataset (10 classes).
2. Fine-tune 1 or 2 layers in two large pre-trained CNNs (VGG16 and ResNet50) on CIFAR-10.
3. Transfer knowledge from the fine-tuned large CNN(s) to the small CNN via knowledge distillation.
4. Evaluate the effectiveness of knowledge distillation from a single teacher and an ensemble of two teachers.

Dataset and Preprocessing

- Dataset used: CIFAR-10 (60,000 images, 10 classes).
- Data split: 50,000 training and 10,000 testing images.
- Preprocessing: Normalized pixel values to $[0,1]$ and applied one-hot encoding to labels.

Model Architectures

Small CNN

- Two Conv2D layers (32 and 64 filters) + MaxPooling.
- Flatten layer followed by Dense layers with 64 neurons and output of 10 neurons (softmax).

Large CNNs

- VGG16 and ResNet50, pre-trained on ImageNet.
- Modified by removing the top layer and adding GlobalAveragePooling, Dense (128 units), and output layer for 10 classes.
- Last 2 layers were fine-tuned.

Training Results

Table 1: Training and validation performance after 10 epochs

Model	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
Small CNN	81.51%	0.5270	68.59%	1.0191
VGG16 (fine-tuned)	69.34%	0.9044	65.73%	0.9376
ResNet50 (fine-tuned)	29.31%	2.0456	29.34%	2.0300

Observation: ResNet50 underperformed significantly compared to VGG16 and the small CNN, possibly due to insufficient fine-tuning or mismatch with input resolution.

Knowledge Distillation

Two variants of distillation were performed to train the small CNN (student):

1. **From VGG16 only (single teacher)**
2. **From ensemble of VGG16 + ResNet50 (two teachers)**

Evaluation on Test Set

Table 2: Test set performance of small CNN models

Model	Test Accuracy	Test Loss
Small CNN (original)	67.55%	1.0191
Small CNN distilled from VGG16	69.50%	0.9376
Small CNN distilled from VGG16 + ResNet50	68.27%	0.9466

Analysis

- The small CNN trained from scratch achieved reasonable accuracy (67.5%) on CIFAR-10.

- Knowledge distillation from VGG16 improved the small CNN's performance to 69.5%, indicating effective transfer of learned representations from the larger model.
- Distillation from the ensemble of VGG16 and ResNet50 did not outperform distillation from VGG16 alone, likely due to the poor performance of ResNet50 as a teacher.
- The validation and test losses decreased after distillation, indicating better generalization.
- The suboptimal ResNet50 fine-tuning suggests further hyperparameter tuning or unfreezing more layers could be beneficial.

Conclusions

- Knowledge distillation is a promising method to improve a small CNN's accuracy without increasing its complexity.
- Single-teacher distillation was more effective than ensemble distillation here, probably because of the weak ResNet50 teacher.

You can find the implementation on: **GitHub Repository**.